VIB-KU LEUVEN
CENTER FOR
CANCER BIOLOGY

Genome editing in mammalian cell models

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VIB-KU Leuven Center for Cancer Biology

21 August 2026

Why editing in mammalian cells at all?

1

Driver or passenger

Sequencing correlates. Only perturbation shows cause.

2

Genome scale

20,000 genes, "one flask", one readout.

3

Isogenic models

Build the lesion into a matched background.

4

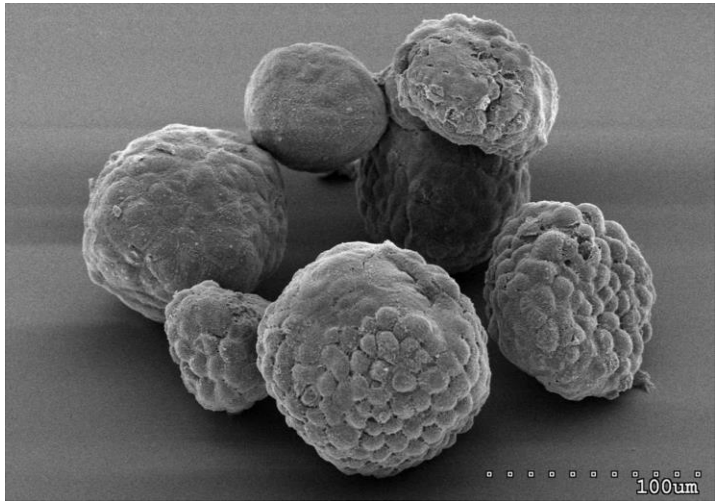
Be the therapy

Since 2023, editing is a registered medicine.

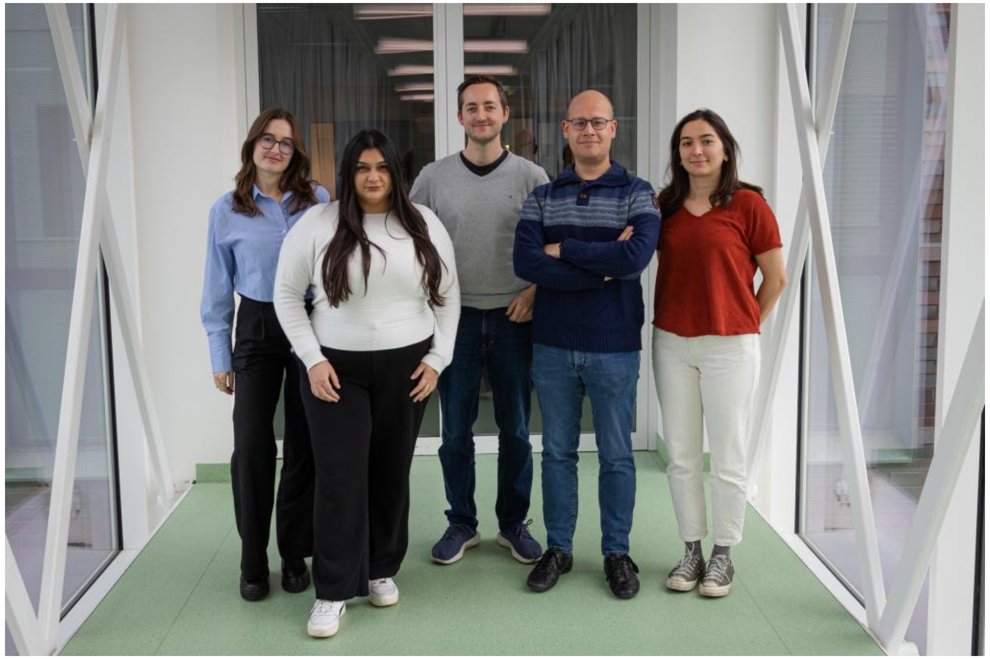
About me:

Genome-wide and developed multiplex CRISPR screens at Cold Spring Harbor,
Now focused on pancreatic cancer at VIB.

- **PhD — Humboldt University and Bayer, Berlin.** Target discovery and epigenetics.
- **Postdoc — Vakoc lab, CSHL.** Paralog co-targeting CRISPR screens.
- **Since 2025 — VIB-KU Leuven.** PDAC plasticity, organoids, in vivo.



EM of cancer cells (PDAC organoids)



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Overview

1

Peculiarities

What is easy and what is hard to do in mammalian cells

2

The pipeline

The decisions that matter most

3

Scale

Pooled screens — what is possible in mammalian cells systems.

Not repeated here

- Repair mechanism, gRNA design, Sanger genotyping.
- Base and prime editing, HDR enhancement.

01
Peculiarities

Your cells are not haploid yeast

Ploidy and local copy number decide what “knockout” means.

Haploid yeast / HAP1

1 copy

One cut. One clone. Done.

Diploid human cell

2 copies

Both alleles, both frameshifts.

Aneuploid cancer line

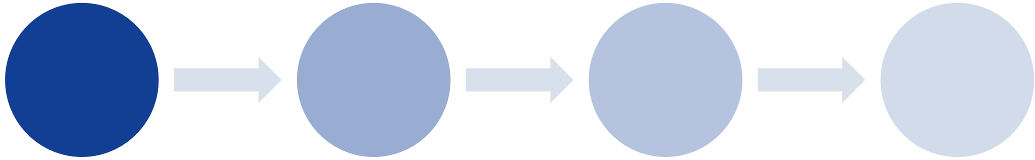
3-many more copies

Every copy, or no phenotype.

An indel on a gel is not a null. One un-cut allele out of four rescues the phenotype.

Look up line ploidy and the copy number at your locus before you design anything — DepMap has both for common lines.

Cancer lines are aneuploid — and they drift



Copy number, mutations and cell state all move. Two labs that “use MIA PaCa-2” are often not using the same cells.

Best practice

- Fingerprint (STR or SNP) every six months.
- Work from one low-passage master stock.
- Re-check copy number at your locus.
- Put the passage number in the figure legend.

“It worked last year” and “it worked in their lab” are sometimes not CRISPR problems.

Editing leads to a population of mutants, not a single mutant

What “90% edited” contains

two frameshifts	**null**
in-frame + frameshift	**hypomorph**
frameshift + wild type	**heterozygous**
unedited	**no edit**

Read the distribution

Amplicon NGS gives allele frequencies.

Report % frameshift alleles, not % “edited”.

Editing leads to a population of mutants, not a single mutant

Discovery of cancer drug targets by CRISPR-Cas9 screening of protein domains

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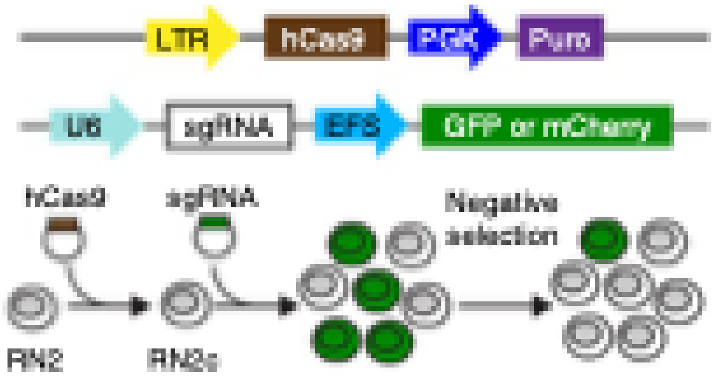

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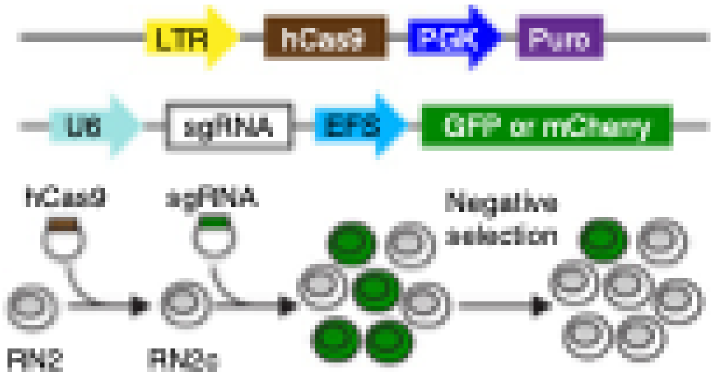

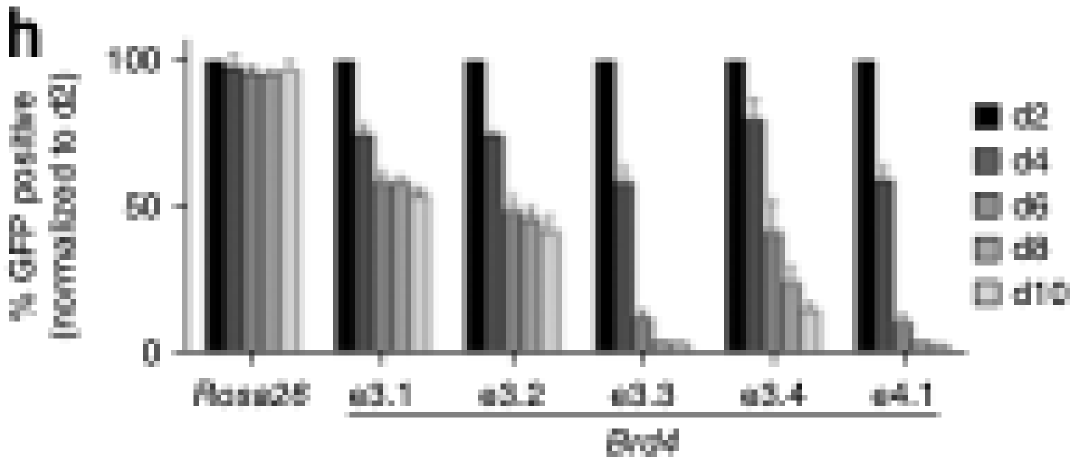

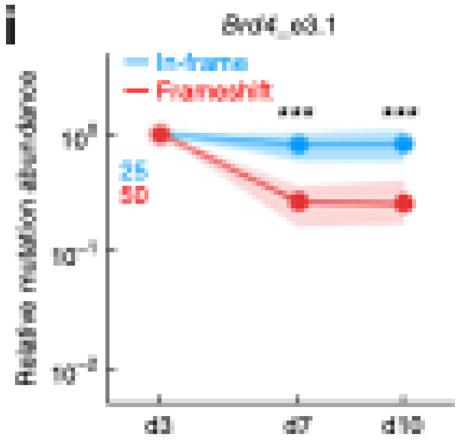

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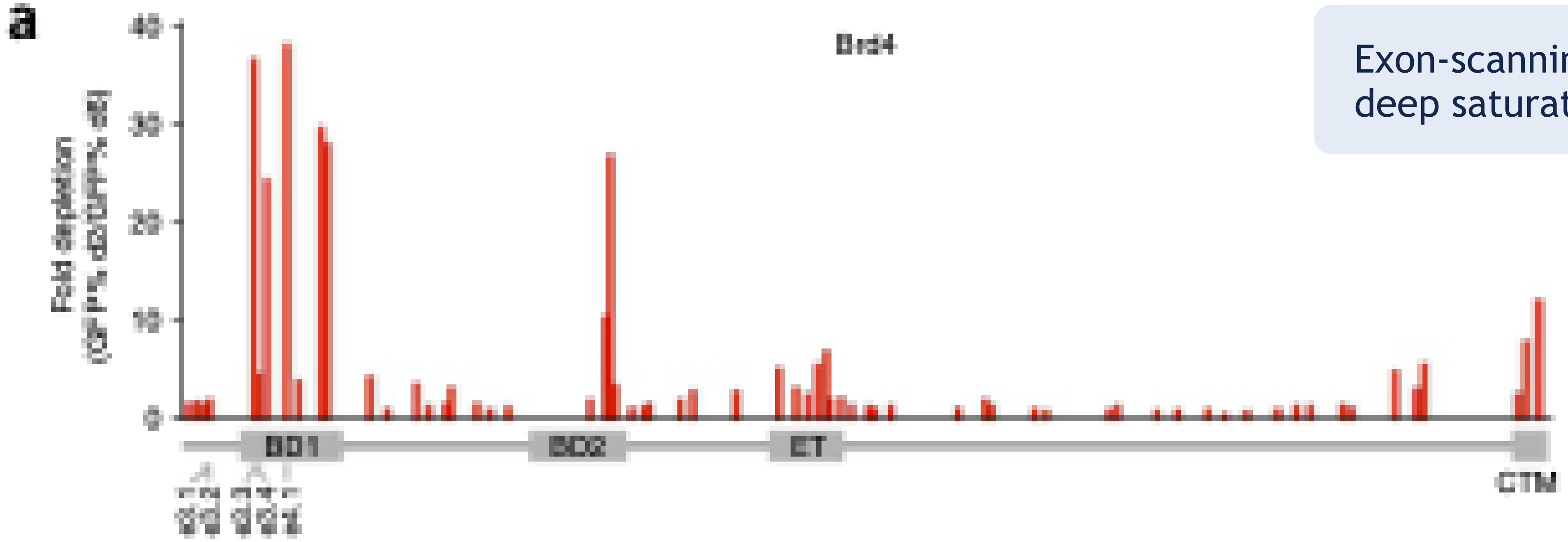
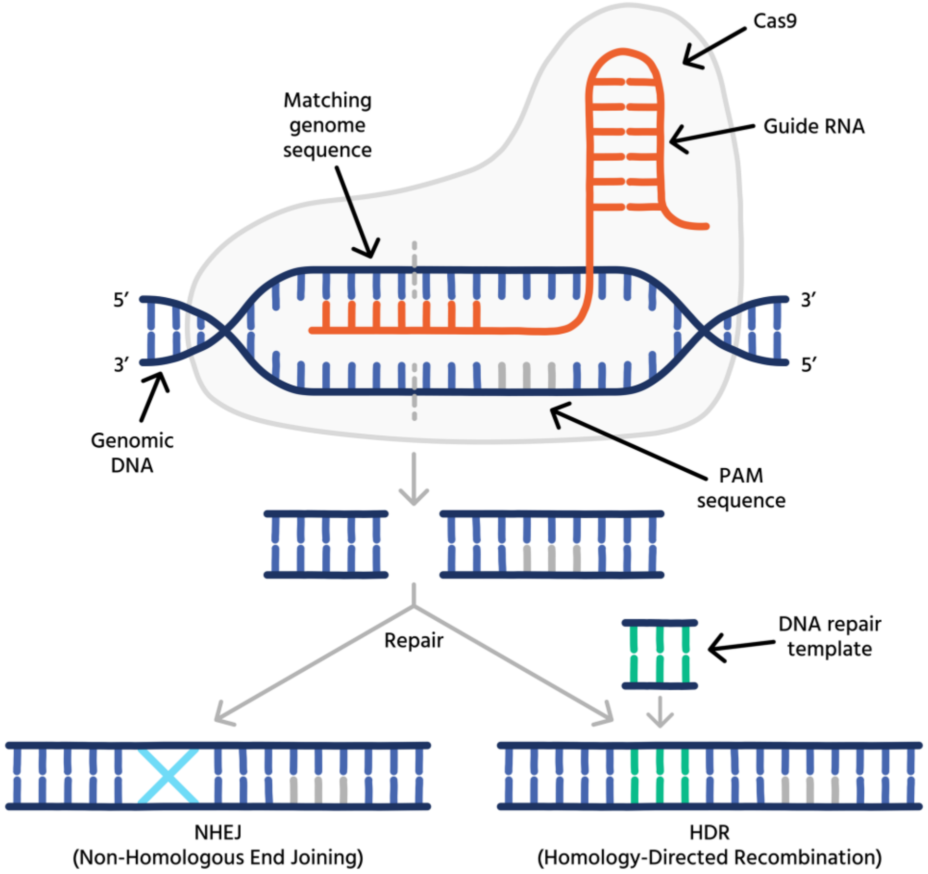


Exon-scanning (simplified version of a deep saturation mutagenesis screen)

NHEJ is common, HDR is rare

- **60-90% indels vs 0.5-5% knock-in.** Two orders of magnitude apart.
- **NHEJ works all cycle.** HDR needs S/G2 and a template nearby.
- **Large deletions and on-target LOH** look like clean knockouts.

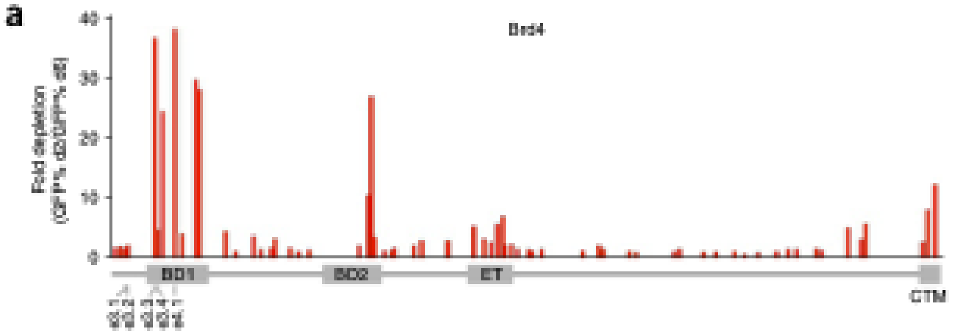
A knock-in project is a screening project. Budget clones, not weeks.



We can select for the vector, not for the edit

In a mammalian cell

Puromycin selects cells that took the vector. The edit stays "invisible". However, we can also use dual sgRNA vectors now.



How do we enrich for editing ? There are four ways:

Co-selection

A surrogate cut site you can select on.

Sort

FACS on the protein or a tagged allele.

Clone

Slow, honest. Genotype every clone.

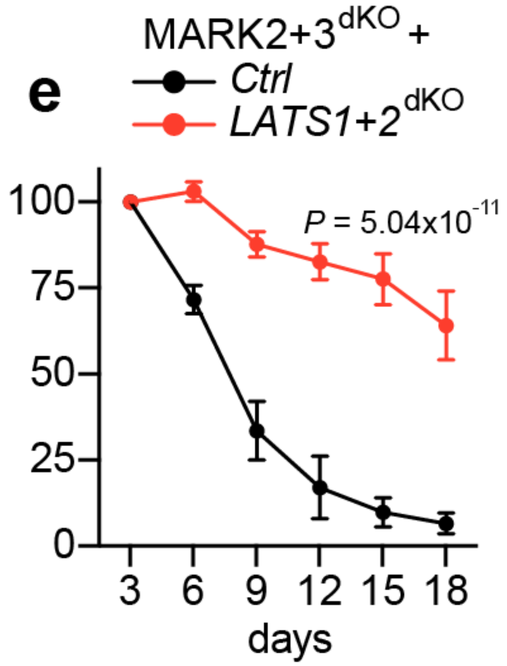
Accept the pool

Valid — if you report the distribution.

How to handle essential genes

- **You will still get clones.** In-frame, hypomorphic, or compensated. Survivorship.
- **The signature:** high editing at day 3, no confirmed clones at day 21.
- **This is how projects quietly stall** — six months of picking clones for a common essential.

What to do? Check, DepMap common-essential flag, effect size in your lineage, and whether it has a paralog.



GFP competition assay: double-knockout cells are gone from the population in ~2 weeks. Klingbeil et al., Cancer Discovery 2024.

When the gene is essential, you can do this instead

Inducible Cas9

Edit when the experiment is ready, not when the clone was made.

hours-days

Degron tag (dTAG)

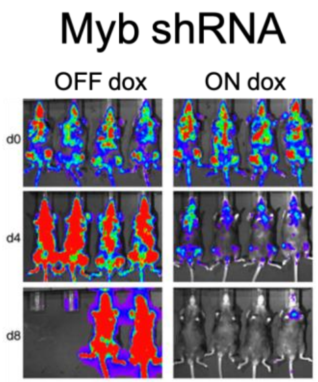
Remove the protein, not the gene. The cleanest primary phenotype.

30-60 min

CRISPRi

Partial and reversible. Models a drug better than a null does.

titratable



Dox-controlled, reversible loss in vivo. Zuber et al. 2011.

Acute is not the same experiment as adapted

Four hours after protein loss vs three weeks in a clone that survived without the gene

The model system is the experiment

throughput ⟵――――――――――――――――――――――――――⟶ *relevance*

2D cell line

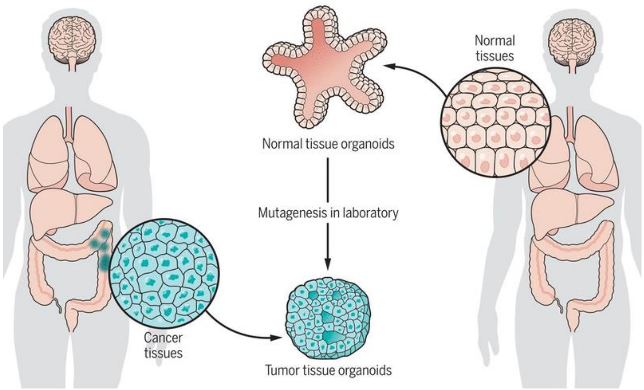
10^8 cells · pooled screens · weeks

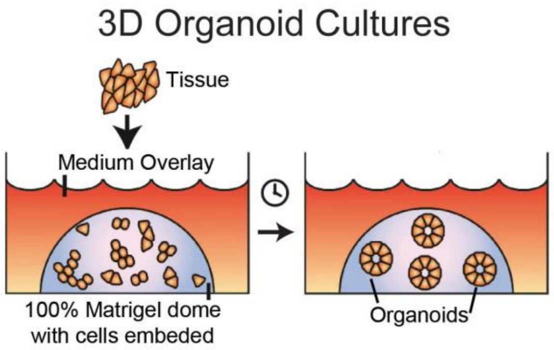
Organoid

10^6 cells · small pools · months

Orthotopic / GEMM

in vivo · AAV library · quarters





Patient and mouse tissue → organoid lines → Matrigel dome culture

02
The pipeline

The pipeline

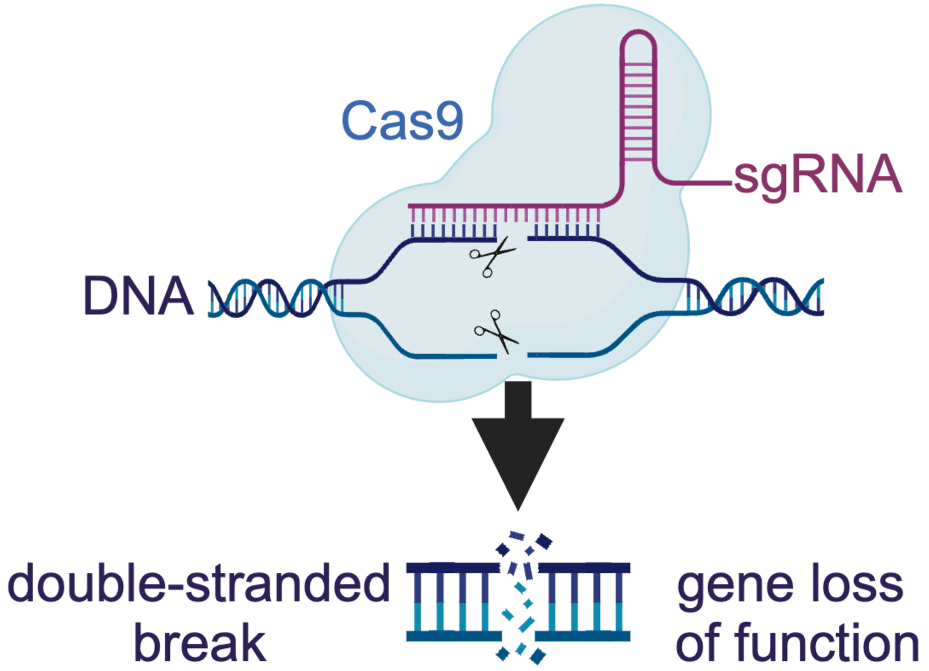
1	2	3	4	5
Design	**Deliver**	**Select**	**Validate**	**Phenotype**
Guides from a validated library	Lentivirus, or RNP electroporation	Antibiotic, sort, or single-cell clone	Amplicon NGS, western, pathway	The actual experiment
3-5 days	1-7days	2-4 wks	2-4 wks	open-ended

Six to ten weeks of setup before you learn anything biological.

Which Cas protein

- **SpCas9 is the default** because the validated libraries and the known failure modes are there. (NGG-3’)
- **PAM-relaxed and high-fidelity variants** buy access or specificity, and cost efficiency.
- **Cas12a** multiplexes from one array — "good" for combinatorial screens. (5'-TTTV)

Moving your target site is almost always cheaper than changing your nuclease.



SpCas9-sgRNA, blunt double-strand break, indel-mediated loss of function

gRNA design in practice: only if you have to

Library	Perturbation	Use it for
Brunello / Brie / Depmap	SpCas9 knockout	Human / mouse genome-wide dropout
TKOv3	SpCas9 knockout	Genome-wide, well-benchmarked controls
Dolcetto	CRISPRi	Essential genes, titratable knockdown
Calabrese	CRISPRa	Gain of function, resistance screens
Bassik / Humagne	Sublibraries	Focused sets: kinases, TFs, druggable

Why ?

measured activity across hundreds of lines.

design tools give you only a prediction.

If you must design your own: two to three guides (dual), early shared exons, and check the paralog.

The off-target worry

On-target site

2 copies, 2 cuts, the edit you designed.

A near-perfect site in a 12-copy amplicon

12 simultaneous cuts in one region → large deletions, loss of heterozygosity, rearrangements.

cuts per cell

Copy number multiplies cut number.

Controls to have:

- Two independent guides, non-overlapping sites, same phenotype.
- A CRISPR-resistant cDNA rescue.
- A non-targeting guide and a known essential.

Rescue beats sequencing. If the cDNA brings the phenotype back, the off-target debate is over.

Delivery: the decision matrix

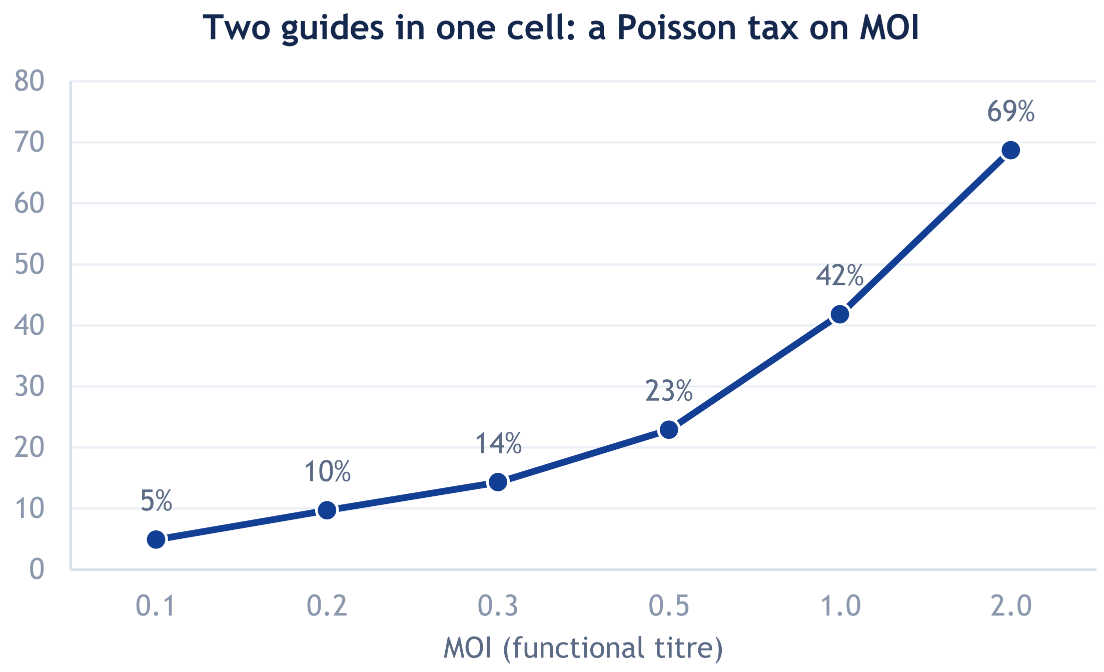
Method	Best for	Genome	Downside
Lentivirus	Most cancer lines; every pooled screen	Integrates	Silencing, MOI, 1-2 weeks
RNP electroporation	Primary cells, iPSC, T cells	Nothing left behind	Cell death, cost per sample, no selection
Plasmid transfection	Easy lines: HEK293, U2OS	Random integration risk	Low efficiency in hard lines, no stable integration (readout)?
AAV / in vivo	Tissue, GEMM screens	Episomal, mostly	Titre, serotype, immune response

Cell type can rules out options

Lentivirus works liable -dose control is key

- **Titre functionally on your cells**
- **Screen at MOI 0.2–0.3.** "Most" transduced cells then carry one guide.
- **Expect silencing** of the cassette over months, and position effects between clones.

High MOI can ruin a screen: multi-guide cells create false interactions and blur effect size.



RNP (ribonucleoprotein) an alternative

How long the nuclease is present

Lentiviral Cas9	weeks to months of continuous expression
RNP	24-48 h, then gone

What you gain

- Highest editing per event in primary cells.
- No cassette to silence.
- Lower total off-target burden.
- No immune reaction - later

What you pay

- No selection marker — sort, clone, or accept a pool.
- Consumables per sample are real.
- Electroporation kills a fraction of the cells.

Where I used it

- Primary and patient-derived cells.
- iPSC and T cells.
- Knock-in with an ssODN donor.

Selection and clone derivation

1
Antibiotic
Selects the vector, not the edit.
5-10 days

2
Sort
FACS on a reporter or the protein.
1 day

3
Single-cell clone
Limiting dilution or a dispenser.
3-5 weeks

4
Genotype and keep
Amplicon NGS. Keep ≥2 per genotype.
1 week

Clonal artefacts;
Two independent clones per genotype, plus a rescue where you can.

Keep the parental pool
Same passage, same medium, same handling — good to have if you want to pick more and compare.

Validation at the protein is best

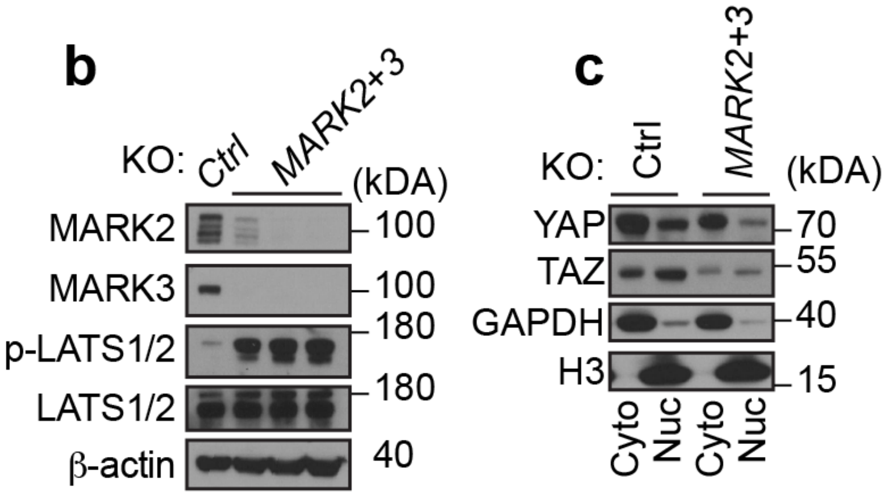
Exon skipping
The cut exon is spliced out, in frame.

Alternative start
Translation re-initiates downstream.

In-frame hypomorph
Reduced activity, not absence.

NMD escape
A dominant-negative fragment survives.

Antibody for the protein, and a readout for the pathway.



MARK2/MARK3 knockout: both proteins gone, and the downstream signal with them. Klingbeil et al., Cancer Discovery 2024.

03
Scale

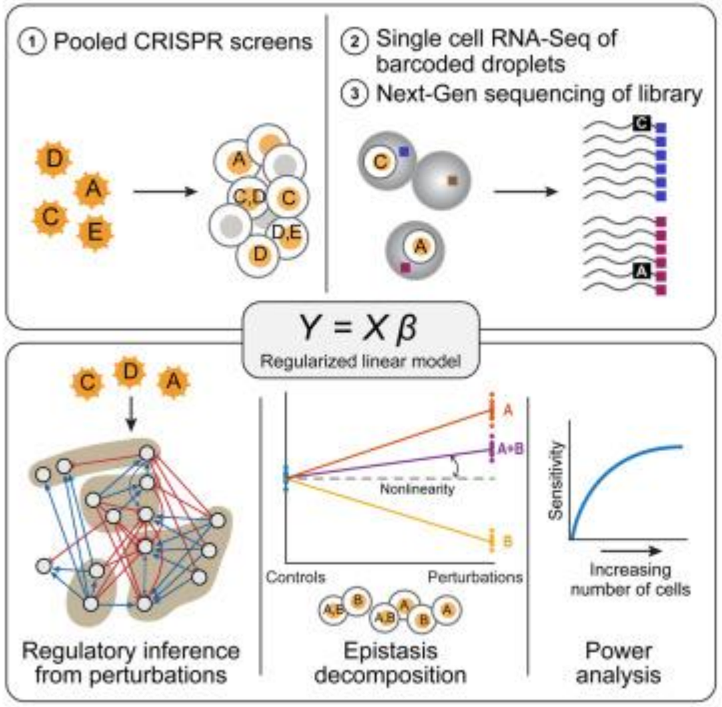
This we can do now in mammalian cells

20,000 genes × 4 guides

in one population, under one selective pressure, read out in one sequencing run.

- **The readout is the flexible part:** survival, a reporter, a marker, a transcriptome, an image.

Best design: A pooled screen to answers one question about 20,000 genes — not 20,000 questions.



Pooled perturbation with a single-cell readout

Anatomy of a pooled screen

Library in at controlled MOI, pressure applied, guide distribution out.

1 — Library
Plasmid pool → virus

2 — Transduce
MOI 0.2-0.3, ≥500 cells/guide

3 — T0 sample
Harvest before pressure

4 — Pressure
14-21 days

5 — Sequence
gDNA → PCR → NGS

The comparison you actually make

$\log_2$(endpoint / T0), collapsed per gene across guides and replicates.

T0 is not optional

Without it you cannot tell a guide that dropped out from one that was never there.

IMPORTANT: have larger number of controls; If your known essentials do not drop out, the screen failed.

Screen design - a statistics problem

≥500×

cells per guide

at every step, including T0 and every passage

≥3

replicate infections

not three PCRs of one infection

≥95%

of the library present

sequence the pool before you start

200-500

reads per guide

the least expensive step of the screen

The narrowest passage defines the noise.

80,000 guides × 500 cells is 40 million cells to carry at every split.

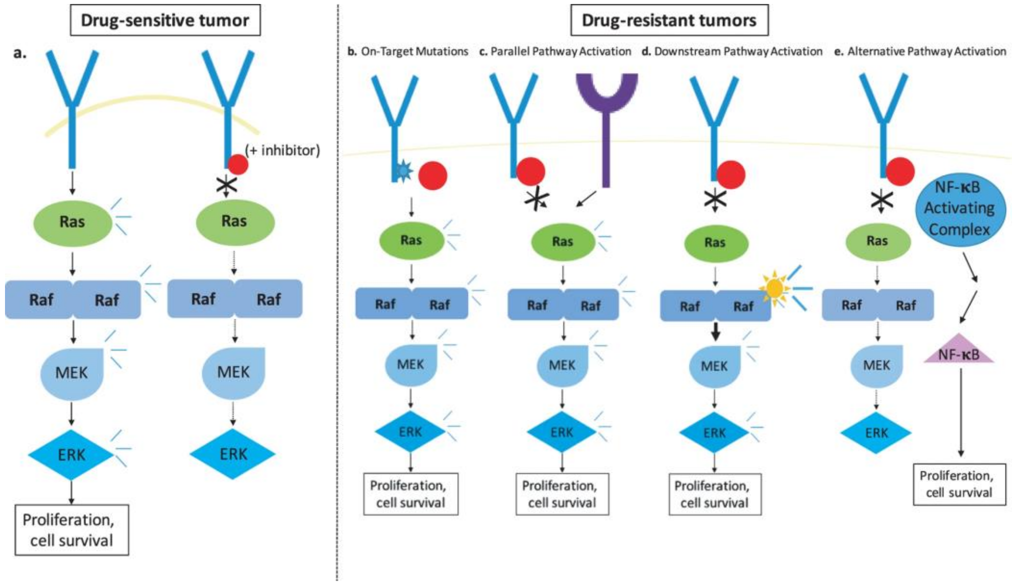
What screens are genuinely good at

Dependencies
What this line needs. Cleanest signal, most crowded field.

Resistance
What survives your drug — the most clinically predictive screen.

Synthetic lethality and paralogs
What is needed only in this genotype.

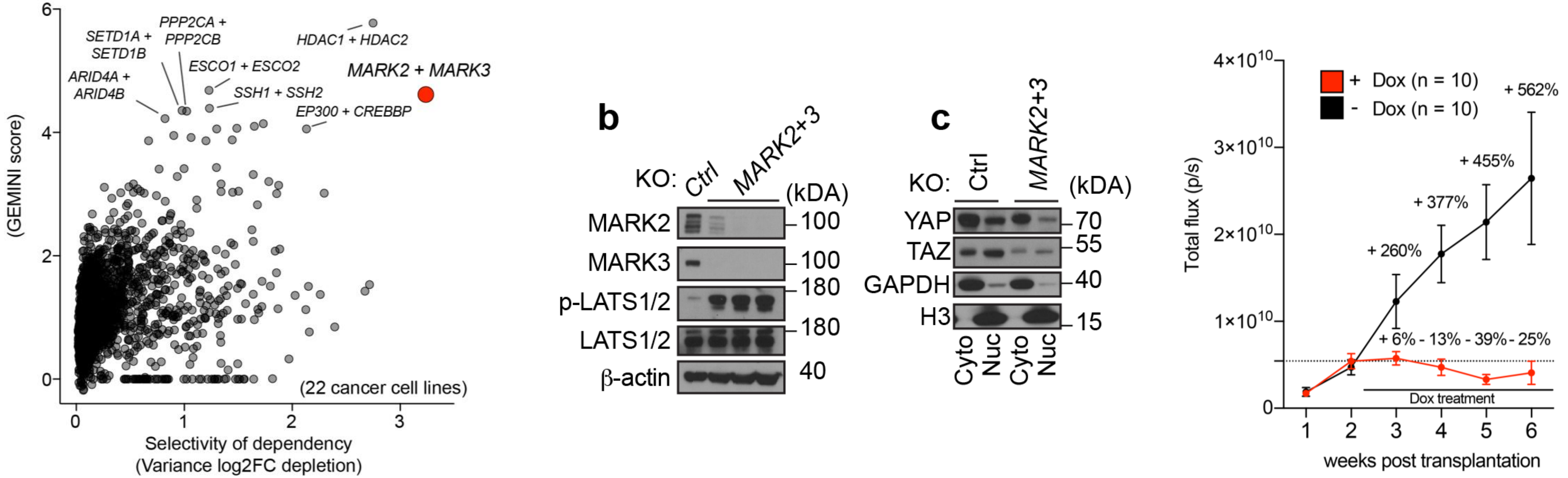
Modifier and marker screens
Anything you can sort on. Escapes the viability bias.



Resistance mechanisms mapped by perturbation under drug pressure

A worked example, end to end

Paralog screen → MARK2/3 → Hippo → orthotopic PDAC. Cancer Discovery 2024.



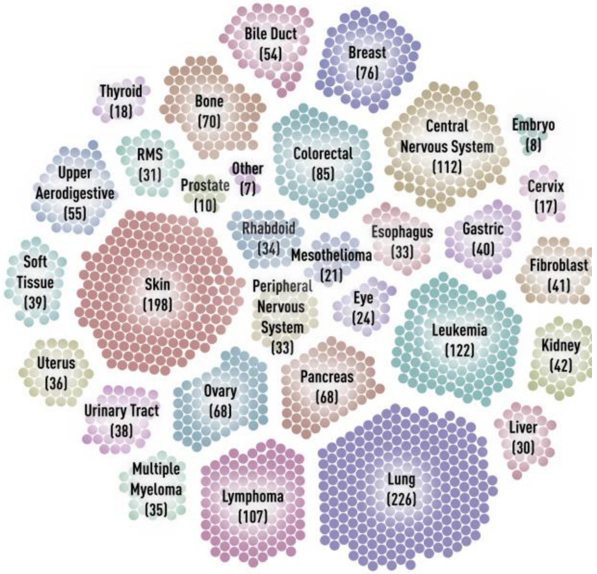
Screen	Hit	Mechanism	Model
>5,000 paralog pairs	MARK2 + MARK3	Hippo switched off	orthotopic PDAC, in vivo

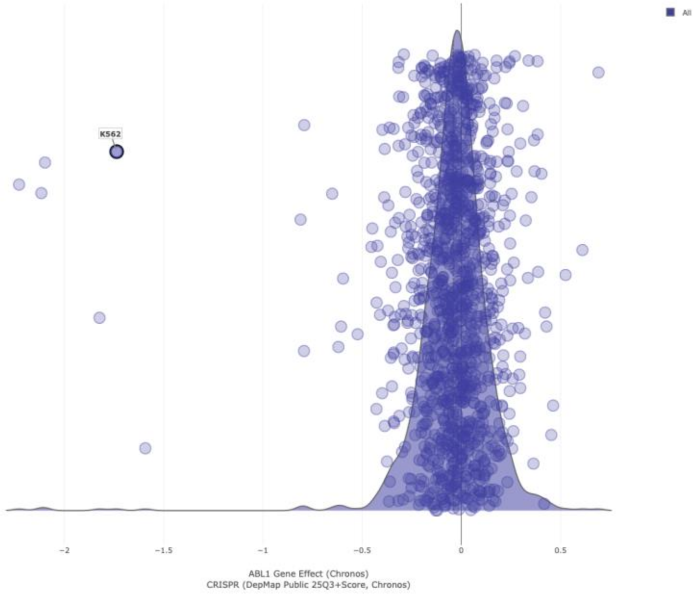
All screens took 4-6 months. Everything after it took two years.

An amazing resource: DepMap

Someone may already have run your experiment.

- Is it a dependency in your lineage?
- Is it a common essential? Then see slide 11.
- Is your line in there — and what is the copy number?
- Does it have a paralog that will mask the phenotype?

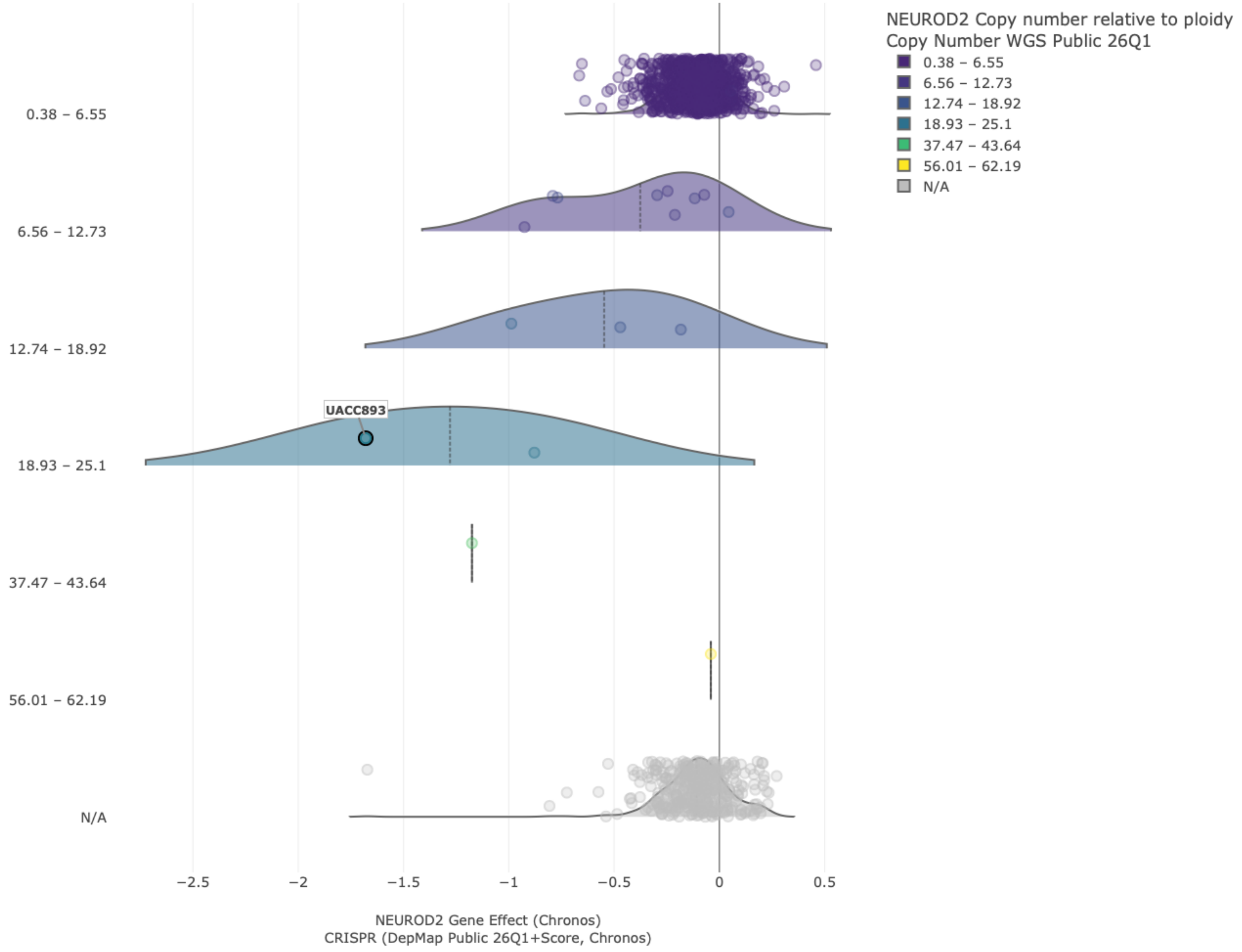




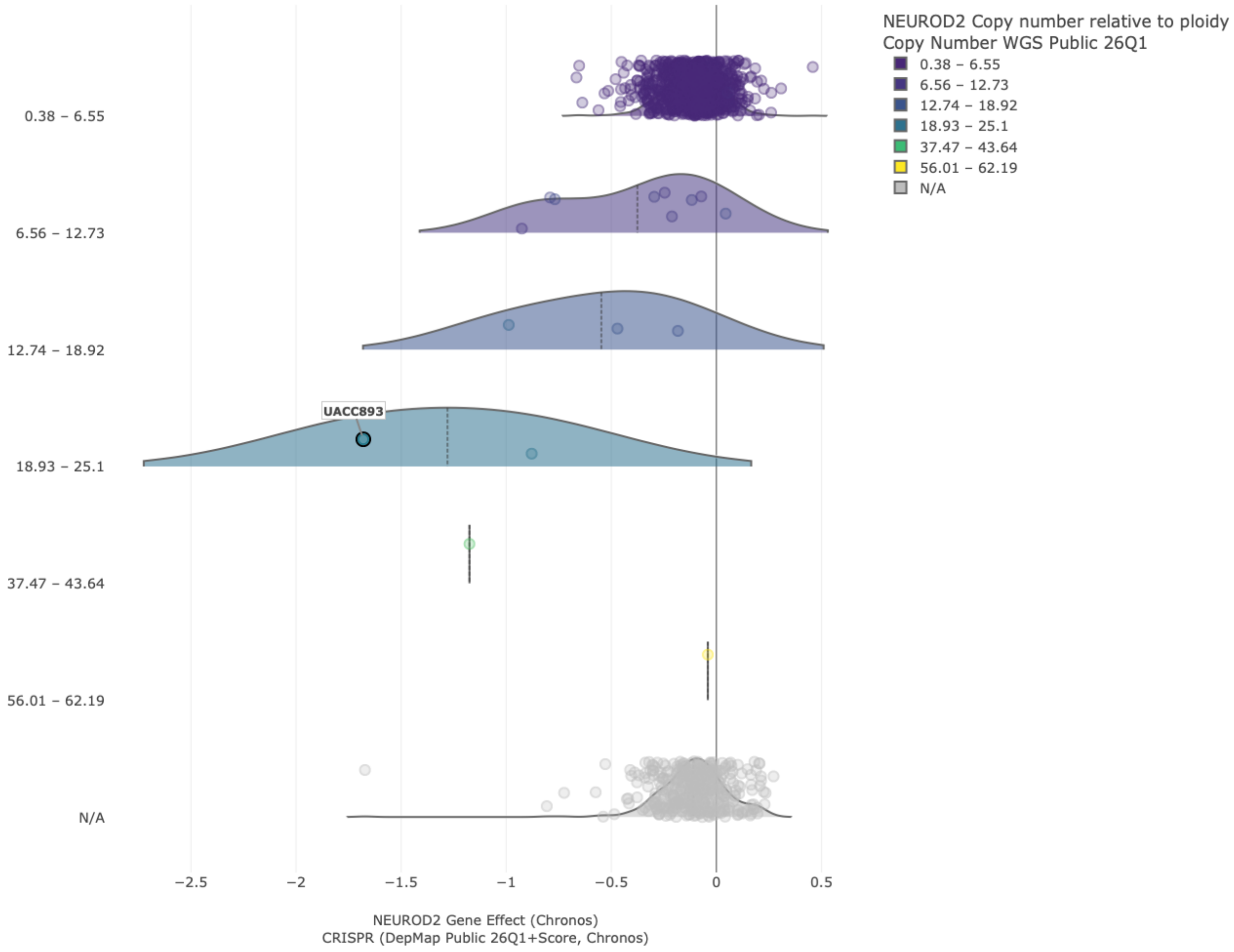
1,100+ cancer lines: genome-wide knockout, expression, mutation and copy number. depmap.org

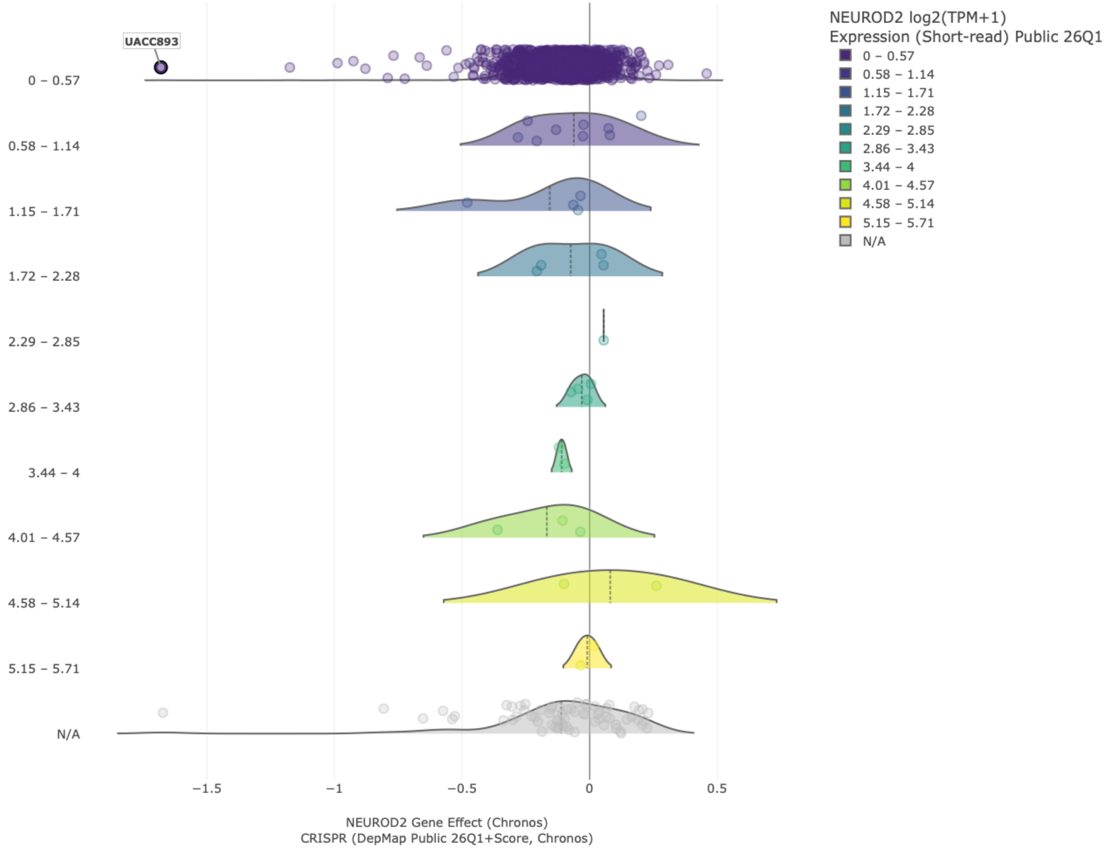
Live demo: depmap.org (example NEUROD2 (CN), SOX9 (off-target))

DepMap NEUROD2

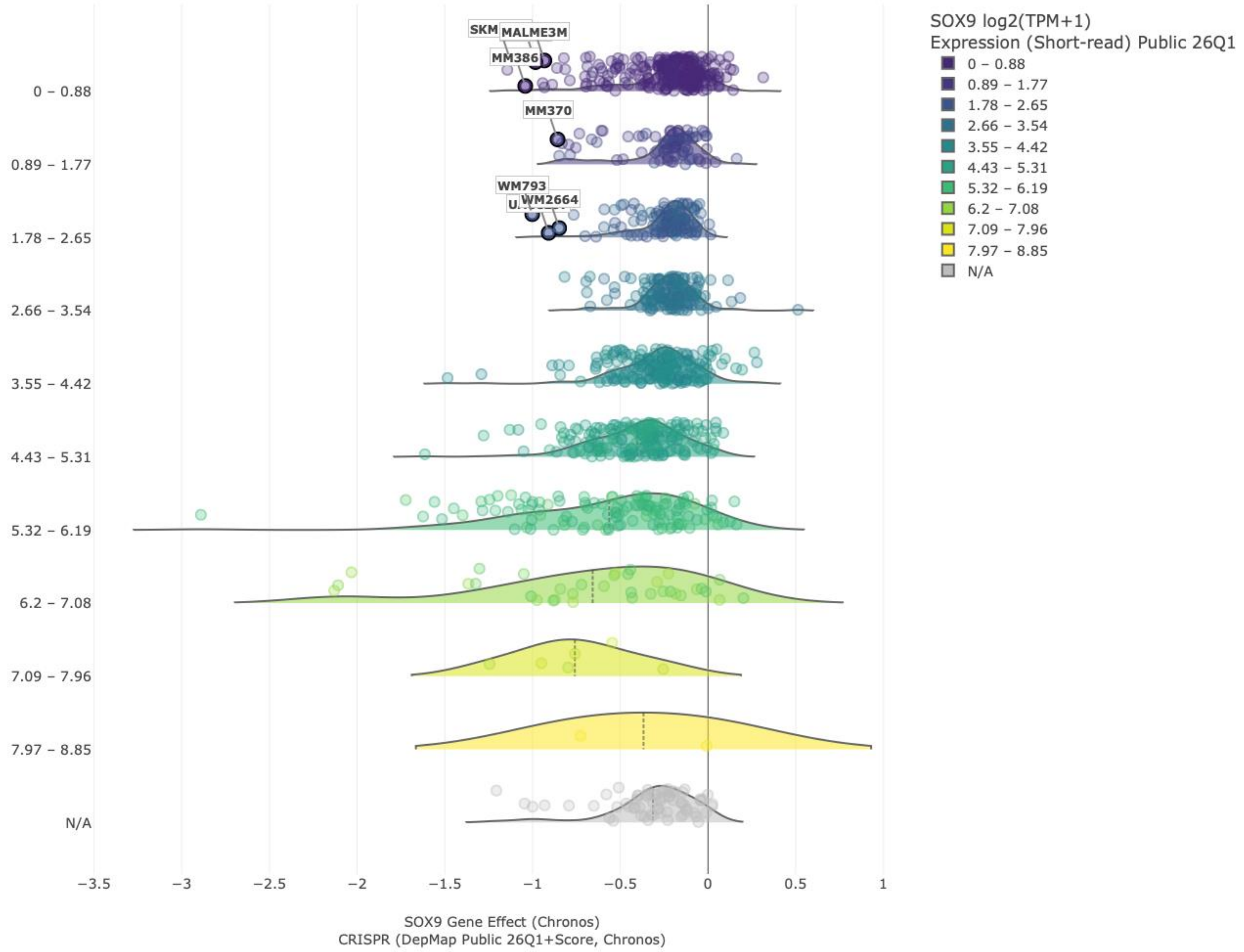


DepMap NEUROD2





DepMap SOX9



SKM
MALME3M
MM386
MM370
WM793
WM2664
0 – 0.88
0.89 – 1.77
1.78 – 2.65
2.66 – 3.54
3.55 – 4.42
4.43 – 5.31
5.32 – 6.19
6.2 – 7.08
7.09 – 7.96
7.97 – 8.85
N/A
−3.5
−3
−2.5
−2
−1.5
−1
−0.5
0
0.5
1
SOX9 Gene Effect (Chronos)
CRISPR (DepMap Public 26Q1+Score, Chronos)
SOX9 log2(TPM+1)
Expression (Short-read) Public 26Q1
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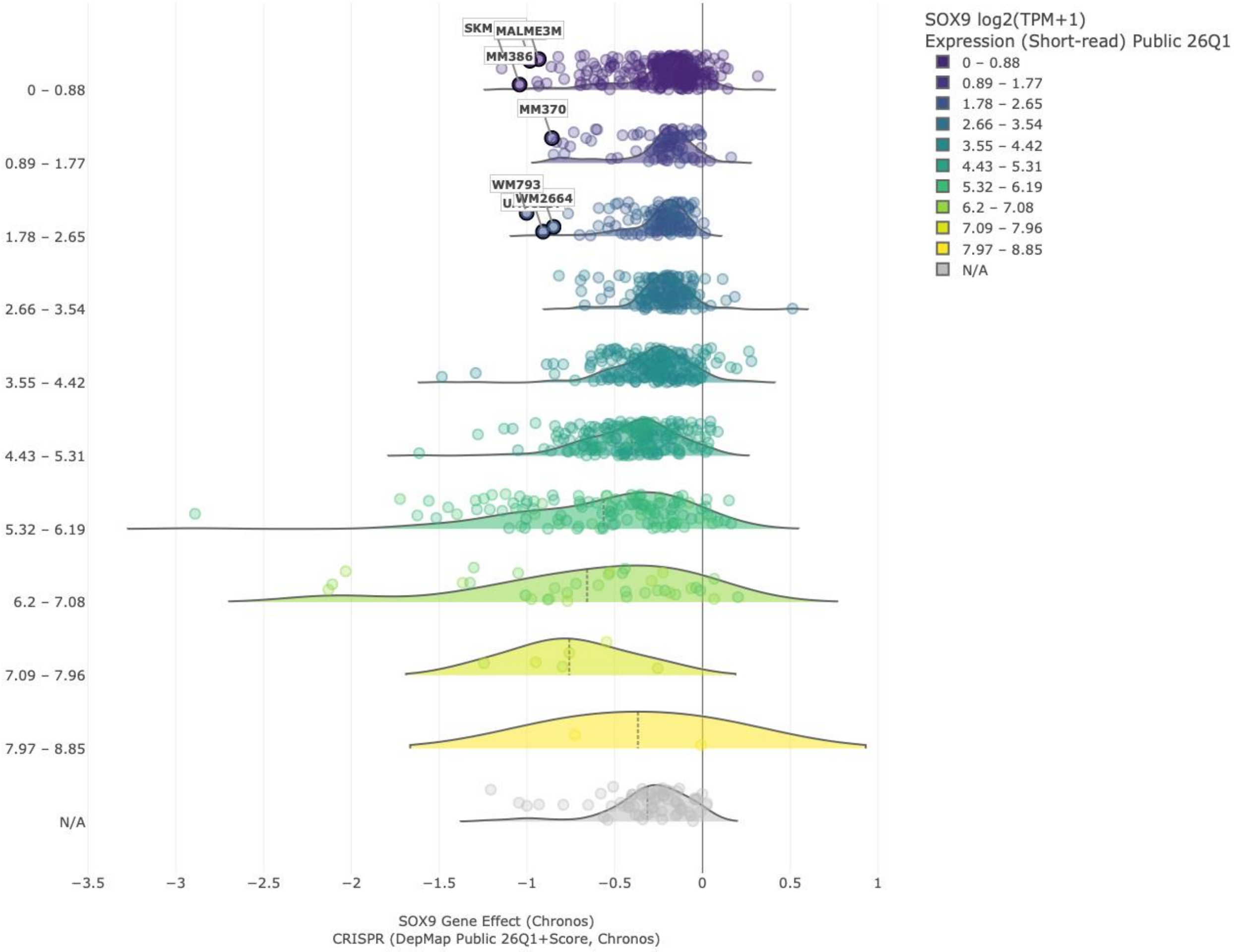

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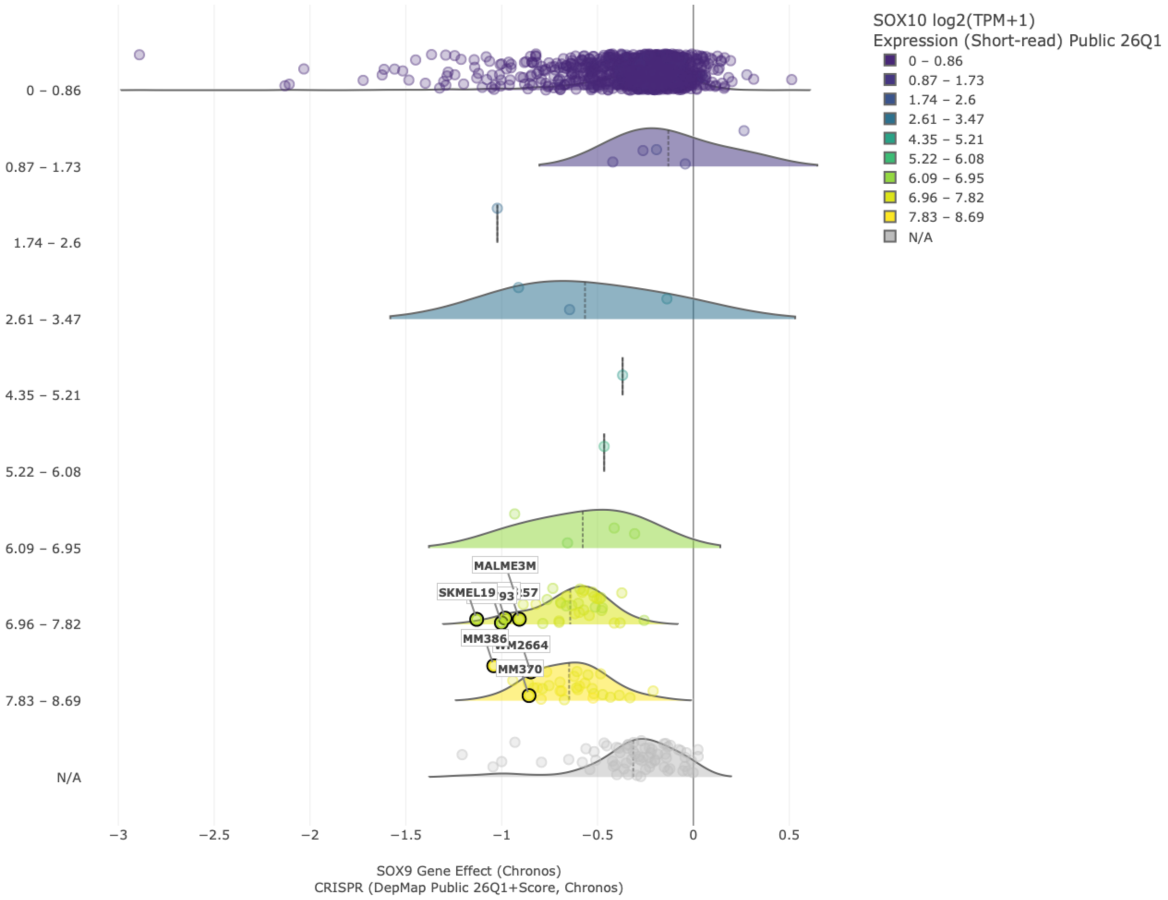

0 – 0.86
0.87 – 1.73
1.74 – 2.6
2.61 – 3.47
4.35 – 5.21
5.22 – 6.08
6.09 – 6.95
6.96 – 7.82
7.83 – 8.69
N/A
MALME3M
SKMEL19
MM386
MM370
−3
−2.5
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SOX9 Gene Effect (Chronos)
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2.61 – 3.47
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6.09 – 6.95
6.96 – 7.82
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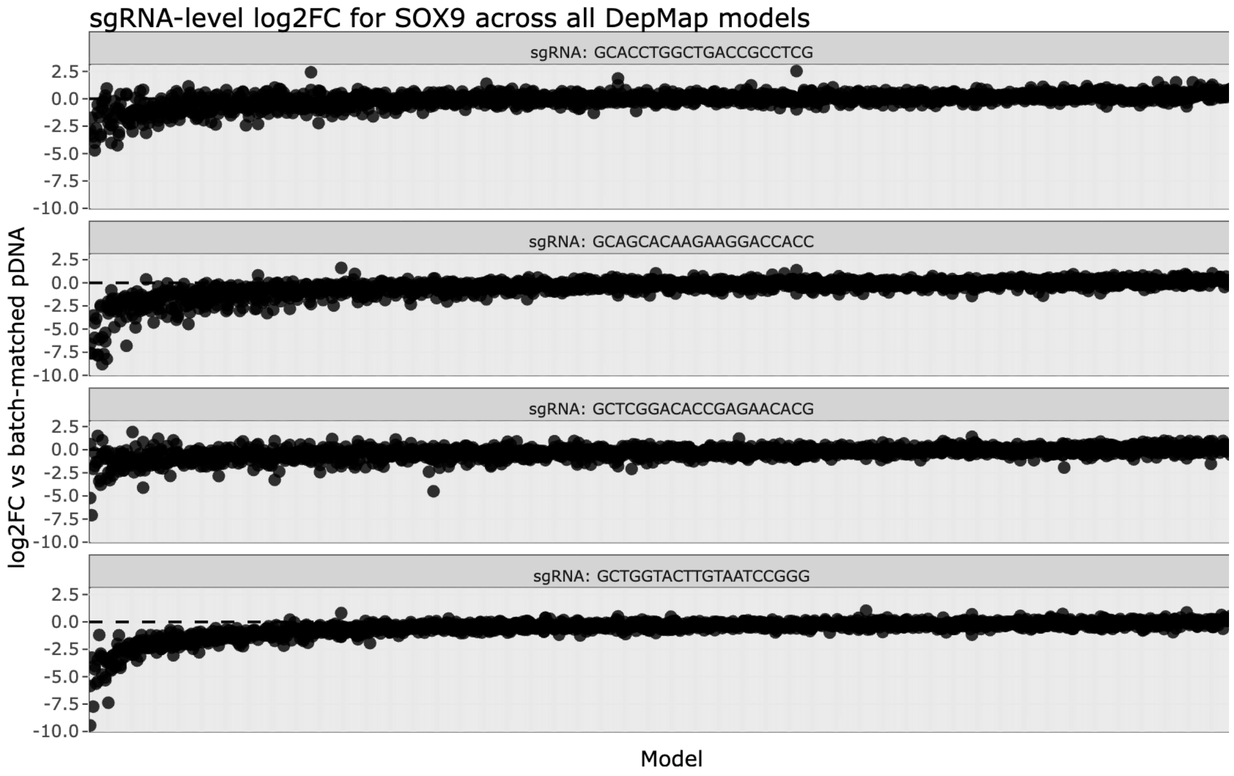
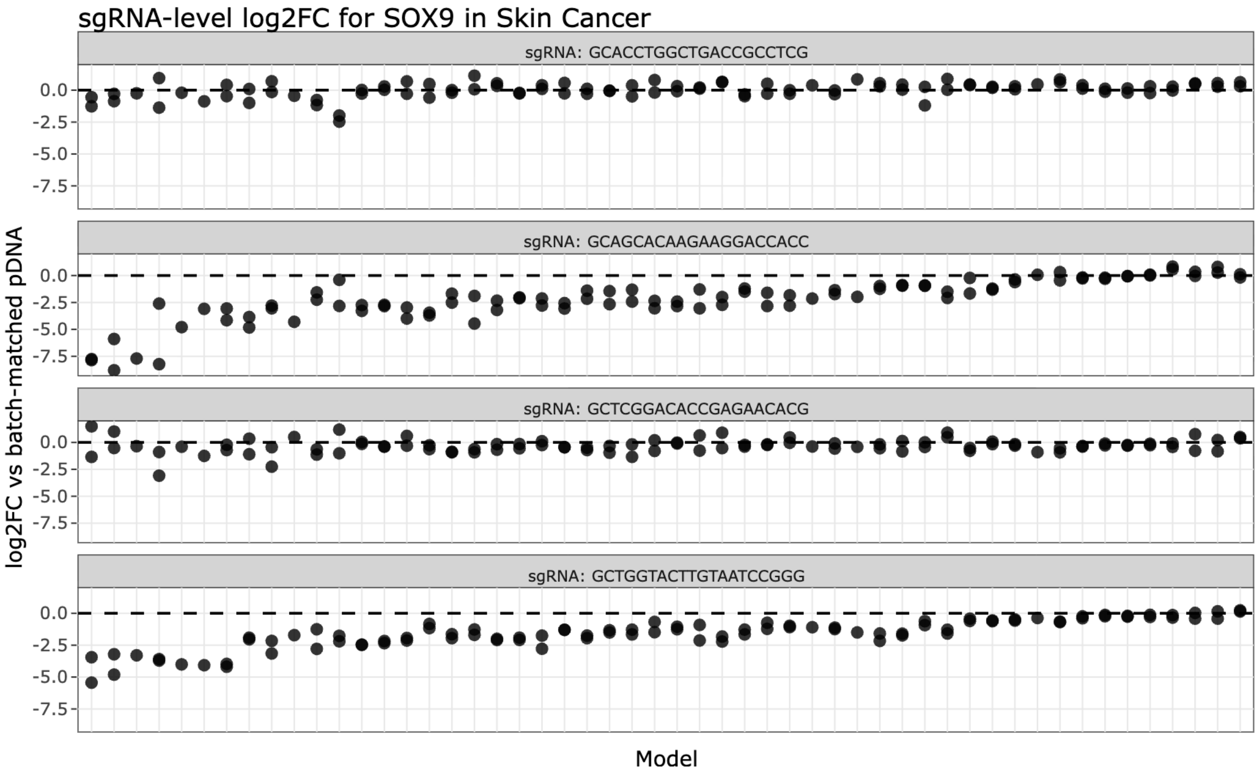
N/A

DepMap SOX9 individual sgRNAs





Beyond cutting, briefly

Base editing

No double-strand break. Choose it to model a specific missense variant.

Prime editing

Small precise edits, no donor. Modest efficiency, moving fast.

CRISPRi / CRISPRa

Titratable, reversible. Choose it for essential genes and dose-response.

Epigenome editing

Silencing without a cut. Best for enhancers and cell state.

All four still depend on delivery, dose, selection and validation at the protein.

Seven things worth writing down

1. Check DepMap and your locus copy number first.
2. Take guides from a validated library. Always two or more.
3. If the gene is essential, go inducible or degron.
4. Titre on your cells. Screen at MOI 0.2–0.3. Keep a T0.
5. Validate at the protein and at the pathway.
6. Keep the parental pool and ≥2 clones per genotype.
7. Write down the passage number. Fingerprint the line.